

NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY
THIRD SEMESTER – B.TECH.
END-SEMESTER THEORY EXAMINATION, JULY 2026

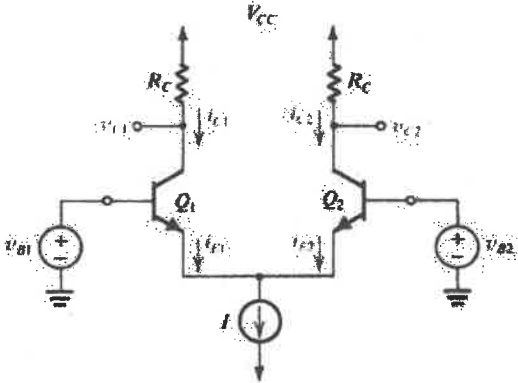
Course Code: ECECC07/VTECC304

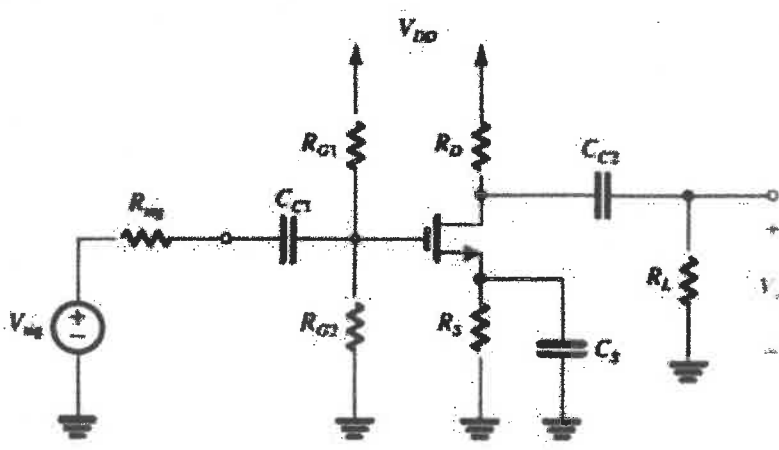
Course Title: Microelectronics

Time: 3:00 Hours

Max. Marks: 50

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	A MOS differential amplifier has tail current = $800\ \mu\text{A}$, $k_n(W/L) = 3.5\ \text{mA/V}^2$, $V_{OV} = 0.25\text{V}$, $R_D = 12\text{k}\Omega$. Determine, (i) differential gain, (iii) output voltage when $v_{id} = 15\text{mV}$.	5	CO1
1b	A Widlar current source uses identical BJTs. Given, $I_{REF} = 2\text{mA}$, $R_E = 470\Omega$, $V_T = 26\text{mV}$. Determine the output current. Write two advantages over a simple current mirror.	5	CO1
1c	An <i>npn</i> differential amplifier shown in Fig.1. with $I = 0.4\ \text{mA}$, $V_{CC}=V_{EE}= 2.5\ \text{V}$, and $R_C= 5\ \text{k}\Omega$ utilizes BJTs with $\beta = 100$ and $v_{BE}= 0.7\ \text{V}$ at $i_C= 1\ \text{mA}$. If $v_{B2}= 0$, find V_E, V_{C1}, and V_{C2} obtained with $v_{B1}=+0.5\ \text{V}$, and with $v_{B1}=-0.5\ \text{V}$. Assume that the current source requires a minimum of 0.3V for proper operation.  Fig. 1.	5	CO1
Q 2	Attempt any 2 parts of the following		
2a	Derive the expression for the small-signal output resistance of a Cascode MOSFET amplifier. Show that the output resistance is significantly higher than that of a Common Source amplifier, and explain the reason for this improvement.	5	CO2
2b	A common-source MOSFET amplifier has transconductance, $g_m = 6\ \text{mS}$, drain resistance, $R_D = 4.7\ \text{k}\Omega$, load resistance, $R_L = 10\ \text{k}\Omega$, gate-drain capacitance, $C_{gd} = 2.5\ \text{pF}$, gate-source capacitance, $C_{gs} = 12\ \text{pF}$, Source resistance, $R_S = 2\ \text{k}\Omega$. Assume the gate current is negligible and ignore all other parasitic capacitances. Using Miller's theorem, determine: (i) The midband voltage gain (A_v), (ii) The equivalent Miller input capacitance.	5	CO2

2c	Using the Open Circuit Time Constant (OCTC) method, derive the expression for the upper cut-off frequency the common-source MOSFET amplifier shown in Fig.2. Clearly state all assumptions made during the derivation.  Fig. 2.	5	CO2
Q 3	Attempt any 2 parts of the following		
3a	Derive the effect of negative feedback on gain, bandwidth, input resistance, output resistance, and noise.	5	CO3
3b	Derive the expressions for the input resistance and output resistance of a Shunt-Series feedback amplifier. Explain the physical reason for the change in resistances due to shunt mixing and series sampling.	5	CO3
3c	A series-series feedback amplifier has open-loop transconductance $G_m = 80 \text{ mS}$, feedback factor $\beta = 0.05$, input resistance $R_i = 2 \text{ k}\Omega$, output resistance $R_o = 20 \text{ k}\Omega$. Determine, (i) Loop gain, (ii) Closed-loop transconductance, (iii) Input resistance after feedback, and (iv) Output resistance after feedback.	5	CO3
Q 4	Attempt any 2 parts of the following		
4a	A Class A amplifier operates from $V_{CC} = 20\text{V}$, $I_{CQ} = 0.5\text{A}$, load resistance $R_L = 10\Omega$. Determine, (i) DC input power, (ii) Maximum AC output power, and (iii) Maximum efficiency.	5	CO4
4b	Explain the operation of a Class B power amplifier using transfer characteristics. Derive the expression for maximum efficiency.	5	CO4
4c	Explain crossover distortion in a push-pull amplifier. How can it be minimized?	5	CO4
Q 5	Attempt any 2 parts of the following		
5a	Explain the operation of the 555 timers in a stable mode with a neat circuit diagram.	5	CO5
5b	Design a 555 astable multivibrator to generate a frequency of 2 kHz with a duty cycle of approximately 60%. Assume a capacitor value of $0.01 \mu\text{F}$. Determine suitable values of R_A and R_B .	5	CO5
5c	Draw the circuit of a Wien bridge oscillator. Derive the expression for frequency of oscillation and amplifier gain.	5	CO5